DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**CRITERIA FOR INTUBATION, EXTUBATION AND WEANING OFF FROM VENTILATOR**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-NICU-017 (v1)**		
Last Version Approval Date	**N/A**	Number of Pages	**10**

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APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
19 / 3 / 2024	7 / 4 / 2024	TRIENNIAL	18 / 3 / 2027

1.0 PURPOSE

1.1. To provide the criteria to physician on when to intubate, extubate and wean-off patient from ventilator.

1.2. To establish guidelines to all healthcare providers on how to intubate and extubate and wean-off from ventilator in a safe and timely manner.

2.0 DEFINITION/ABBREVIATION

2.1. **Intubation**- is the placement of an endotracheal tube in the trachea for the establishment and maintenance of airway.

2.2. **Extubation**- is the process of removing an endotracheal tube from the patient when it is no longer required.

2.3. **Weaning from ventilator**- defined as the process of abruptly or gradually withdrawing ventilator support of the patient.

2.4. **SBT (Spontaneous Breathing Trial)**- assesses the patient's ability to breathe while receiving minimal or no ventilator support.

2.5. **CPAP (Continuous Positive Airway Pressure)**- is a treatment that uses mild air pressure to keep the breathing airways open.

2.6. **T- Piece**. is a T-shaped tubing connected to an endotracheal tube used to deliver oxygen therapy to an intubated patient who does not require mechanical ventilation.

2.7. **ETT (Endotracheal Tube)**- is a tube inserted into the trachea via the mouth or nose to maintain an airway and allow for mechanical ventilation or surfactant administration.

2.8. **RT**- Respiratory Therapist

2.9. **PEEP**- Positive End-Expiratory Pressure

2.10. **PIP**- Peak Inspiratory Pressure

2.11. **VR**- Ventilator Rate

2.12. **SIMV**- Synchronized Intermittent Mandatory Ventilation

2.13. **HFOV**- High frequency

2.14. **Ti**- Inspiratory Time

2.15. **Vt**- Tidal volume

2.16. **PPE**- Personal Protective Equipment

2.17. **ER**-Emergency Room

2.18. **OPD**- Out-Patient Department

2.19. **NICU**- Neonatal Intensive Care Unit

3.0 RESPONSIBILITY

3.1. **Head of Pediatric and Neonatology**

3.1.1. Responsible for ensuring that the criteria for patient's intubation, extubation and weaning off from ventilator are met.

3.2. **Physicians (Residents, Specialist, Consultant)**

3.2.1. Responsible to assess patients that needs to be intubated, extubated or weaned-off from ventilator according to criteria.

3.2.2. Provide a complete written order of the procedure to be done to the patient.

3.2.3. Perform the procedure with the assistance of a trained staff nurse or respiratory therapist.

3.3. Respiratory Therapist

3.3.1. Assist the physician in validating the criteria before the performance of the intubation, extubation and weaning-off procedure.

3.3.2. Aid the physician during the procedure.

3.4. **NICU Nurse (Head nurse, Charge Nurse, Staff nurse)**

3.4.1. Responsible for assisting the physician in performing the procedure to the patient in the absence of the Respiratory Therapist.

3.4.2. Do monitor patient before, during and after the intubation, extubation and weaning off procedure.

3.4.3. Document the procedure in the patient's medical file

4.0 POLICY

4.1. **INTUBATION:**

4.1.1. The criteria for intubation must be followed:

4.1.1.1. For maintenance of a patent airway/ upper airway obstructions

4.1.1.2. For worsening respiratory distress/respiratory failure

4.1.1.3. Prolonged apnea

4.1.1.4. Inadequate ventilation

4.1.1.5. Worsening hypoxia, despite oxygen therapy

4.1.1.6. Extreme prematurity

4.1.1.7. Severe meconium aspiration syndrome

4.1.1.8. Diaphragmatic hernia

4.1.1.9. Congenital/structural airway abnormalities

4.1.1.10. Administration of surfactant

4.1.1.11. Resuscitation

4.2. Appropriate blade size to be used for Endotracheal Intubation:

4.2.1. Extremely Preterm - No.00

4.2.2. Preterm- No.0

4.2.3. Term- No. l

4.2.4. Infant (6-l I kg)- No. l

4.2.5. Small child (12-14 kg)- No.2

4.2.6. Child (l5-18 kg)- No.2

4.3. Recommended ET tube depth of insertion:

4.3.1. Oral Tube Length (cm) = 6 + weight (kg)

4.3.2. Nasal Tube Length (cm) = 6 + (1.5 X weight)

4.4. Endotracheal tube size to be used for Endotracheal Intubation:

4.4.1. For newborn patient:

WEIGHT	GESTATIONAL AGE (weeks)	ET TUBE SIZE (mm)
Below 1000 g	Below 28	2.5
1000 – 2000 g	28-34	3.0
2000 – 3000 g	34-38	3.5
Above 3000 g	Above 38	3.5 – 4.0

4.4.2. For pediatric patient:

WEIGHT	AGE	ET TUBE SIZE (mm)
5 kg	3 months	3.5
8 kg	6 months	4.0
10 kg	1 year	4.0
13 kg	2 years	4.5
17 kg	5 years	4.5
30 kg	10 years	6.0
40 kg	13 years	7.0

4.5. EXTUBATION:

4.5.1. The indications for extubation must be observed:

4.5.1.1. Arteria/venous blood gases are within normal or acceptable limits

4.5.1.2. Spontaneous respiratory effort above the set ventilator rate

4.5.1.3. All vital signs, including oxygen saturations are satisfactory with minimal supplemental oxygen requirement

4.5.1.4. Surfactant administration has been completed

4.5.1.5. With ventilator setting as follows:

4.5.1.5.1. Fio2 below 40%

4.5.1.5.2. Infant weighing above 2 kg with VR 20-25 cycles/minute and pressure 20/3 or less.

4.5.1.5.3. In very small premature babies, VR is l5-20 cycles/minute and pressure 15/3 and sustaining adequate respiration.

4.5.1.5.4. Ti is 0.3-0.4 second

4.6. **WEANING OFF FROM VENTILATOR**:

4.6.1. Indications for weaning of the patient from ventilator are as follow:

4.6.1.1. Patient's primary problem causing need for ventilation has been resolved

4.6.1.2. Patient is awake and responsive

4.6.1.3. Ideally has functioning bowels, absence of abdominal distension

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4.6.1.4. Normalizing metabolic status

4.6.1.5. Adequate hemoglobin concentration

4.6.2. Weaning from different ventilator setting:

4.6.2.1. Weaning from Assist-Control (PTV)

4.6.2.1.1. Decrease PIP but provide adequate Vt (above 4 ml/kg)

4.6.2.1.2. Decrease back up rate 25-30 per minute

4.6.2.1.3. May increase trigger sensitivity to condition respiratory muscles

4.6.2.1.4. Extubate directly from assist-control or switch to SIMV

4.6.2.2. Weaning from SIMV

4.6.2.2.1. Decrease PIP but provide adequate Vt (4 nL/kg)

4.6.2.2.2. Decrease the rate

4.6.2.2.3. Extubate when stable at low rate (i.e. l5 bpm) or combine with PSV.

4.6.2.3. Weaning from PSV

4.6.2.3.1. Adjust level of PIP in PS mode to give adequate Vt (above 4 ml/kg).

4.6.2.3.2. Extubate when stable, spontaneously breathing (no apnea) at low pressure.

4.6.2.3.3. No need to use ETT CPAP pre-extubation

4.6.2.4. Weaning from HFOV

4.6.2.4.1. If low CO2, decrease Amplitude or may increase the frequency

4.6.2.4.2. If well oxygenated, decrease FiO2 until below than 50 % then decrease MAP.

4.6.2.4.3. If MAP l0-12 cm H2O, shift to CMV.

4.6.2.4.4. If MAP 6-8 cm H2O and amplitude less than 20 cm H_2O, then extubate to head box

4.6.3. The needed equipment must be at bedside such as:

4.6.3.1. CPAP machine

4.6.3.2. Oxygen mask

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4.6.3.3. Oxygen cannula

4.6.3.4. Oxyhoods

4.6.4. The weaning procedure must start with the physician's assessment of the newborn's ability for spontaneous breathing. Three main strategies are used to perform SBT which are:

4.6.4.1. T-piece trial, in which only supplemental oxygen is supplied through a T-piece connected to an endotracheal tube.

4.6.4.2. Continuous Positive Airway Pressure (CPAP) trial using a CPAP level equal to the Previous PEEP level.

4.6.4.3. Invasive ventilation with low level of pressure support (5-8 cm H20) or automatic tube compensation.

4.6.5. SBT Duration

4.6.5.1. The physician should wait at least 30 minutes to judge the tolerance of the SBT, but they shouldn't wait longer than 120 minutes if tolerance of SBT is not clear.

4.6.5.2. The initial few minutes of the SBT should be monitored closely before judgment is made to continue the SBT.

4.7. A qualified and competent physician must perform the intubation and extubation procedure with the assistance of at least two (2) qualified nurse and/or RT

4.8. All patients receiving ventilator support should be assessed on a daily basis for their suitability for weaning off from the ventilator.

4.9. All equipment necessary for Endotracheal Intubation, Extubation, and weaning off from ventilator and Emergency drugs must be readily available at all times in the unit

4.10. Alarms of the cardiorespiratory monitor will be attached to new born patient and should be always left on and set at appropriate limits, consistent with the patient's clinical condition and age

4.11. Intubated and ventilated patients must not be left unattended. Lf the bedside nurse should leave. She must inform the charge nurse or another staff nurse to take over the patient.

5.0 PROCEDURE (S)

5.1. **Intubation**:

5.1.1. The Physician will assess the patient's clinical condition and determine the necessity for intubation according to the criteria.

5.1.2. Verify the physician's order and gather all the necessary equipment needed.

5.1.3. Identify the patient correctly.

5.1.4. Perform hand hygiene and wear appropriate PPE as needed.

5.1.5. Place the patient in the pre-warmed radiant warmer and connect to cardio-respiratory monitor.

5.1.6. Position the patient in the sniffing position with rolled towel under shoulders if necessary.

5.1.7. Assist the physician with the intubation procedure.

5.1.7.1. Have the suction ready if the intubator request for it after visualizing the vocal cords and prior to inserting the endotracheal tube

5.1.7.2. Hold the selected endotracheal tube that is still in the package, but with the top of the tube available for the physician to easily grasp.

5.1.8. The physician will intubate the new born.

5.1.8.1. The laryngoscope blade will be passed gently alongside of the mouth and gently pull tongue and epiglottis forward by lifting the blade

5.1.8.2. If the vocal cords and epiglottis do not come into view, pull the laryngoscope back gradually until the cords are visualized.

5.1.9. Ventilate the patient by connecting the ambubag to the inserted endotracheal tube.

5.1.10. With the stethoscope, auscultate the chest for bilateral and equal breath sounds to confirm that the tube is in place.

5.1.11. Secure the endotracheal tube in place by adhesive tapes.

5.1.12. Obtain chest X-ray for confirmation of the ET tube placement

5.1.13. Connect the patient to mechanical ventilator, or proceed to surfactant administration or as directed by the Physician.

5.1.14. Document the procedure done in the patient's medical file

5.2. **Extubation**:

5.2.1. The Physician will assess the patient prior to extubation based on the criteria and the newborn's clinical condition.

5.2.2. Verify physician's order and gather all the necessary equipment needed at bedside.

5.2.3. Identify the patient correctly and ensure that newborn is properly attached to cardiorespiratory monitor.

5.2.4. Perform hand hygiene and wear the appropriate PPE.

5.2.5. Suction ETT and oropharynx five minutes before procedure and ensure that vital signs have returned to baseline prior to extubation.

5.2.6. Place the bed flat and position the patient's head in the midline and slightly extend to promote a clear airway.

5.2.7. Assist the physician to carefully remove the tapes and withdraw the ETT.

5.2.8. Immediately apply the oxygen support that is indicated for the patient. (eg. CPAP, face mask, etc.)

5.2.9. Monitor the patient's vital signs closely and observe for any signs of deterioration.

5.2.10. Document the procedure done in the patient's medical file.

5.3. **Weaning -off from ventilator:**

5.3.1. The Physician will assess the patient's readiness to be weaned-off from ventilator according to the specified criteria.

5.3.2. The nurse will verify the physician's written order for weaning-off and take note of the weaning strategy to be used for the patient. (T-piece, CPAP etc.)

5.3.3. Prepare all the equipment needed for weaning at the patient's bedside

5.3.4. Identify the patient correctly and perform hand hygiene

5.3.5. Assist the physician to wean the patient from ventilator according to policy and written physician's order.

5.3.6. Monitor the patient closely for 30 minutes until the patient tolerates the weaning procedure and until stable

5.3.7. If signs of deterioration are noted, notify the physician immediately and resume to previous ventilatory support or as directed by the physician

5.3.8. If weaning was successful, document the procedure and the patient's response to weaning in the medical file.

6.0 EQUIPMENTS

6.1. Intubation set
6.2. ET Tube
6.3. Suction
6.4. PPE

7.0 ATTACHMENT/FORMS

7.1. Nurses Progress Notes (GDOH-NUR-NPN-203)
7.2. Physician Order Form (GDOH-INP-POF-079)

8.0 REFERENCES

8.1. MOH-KSA Guidelines for Neonatal Care

8.2. https:llacademic.oup.com/bjaed/article/Weaning-from-mechanical-ventilation

8.3. https://www.ncbi.nlm.nih.gov/pmc/articles

8.4. CBAHI Standards NICU 3'd version 7.5. Previous PMSH policies and procedure,20lT revision

9.0 REVISION CLAUSE

9.1. Not Applicable

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وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

PHYSICIAN ORDER FORM

ALLERGIC TO:

DATE	TIME	PHYSICIAN ORDERS	PHYSICIAN STAMP / SIGNATURE	RN INITIAL / JOB NUMBER